

SHINGAN: A Hybrid Opening Range Breakout Framework Combining Zarattini Momentum, Valkyrie Volatility Normalization, and ORBPLUS Execution

Shingan (Clear-Eyed Lens) • Kagemusha (Shadow Warrior) • Hourly Liquidity Lab

Hourly Liquidity Lab, Desktop Workspace, America/New_York • shingan@hourly-liquidity.lab
Code: models/hybrid_orb.py:4 • Data: ict_mechanical/data/intraday_5m (2018–2026)

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Abstract — We present **Shingan**, a hybrid intraday Opening Range Breakout (ORB) strategy that unifies three independently documented edges: **Zarattini et al. (2024) 5-minute abnormal-volume momentum**, **Valkyrie 15-minute volatility normalization (RVOL and OR/ATR phi)**, and **ORBPLUS retracement execution**. Tested on a 44-ticker liquid US universe (2018–2026, 5-minute bars, 1% fixed-fractional risk, \$0.005/share commission, 0.01% slippage, Top-5 daily selection by RVOL), Shingan delivers a 5-year concatenated out-of-sample Sharpe of **1.21** on 1,262 trades (PF 1.25, +108.7% total, MaxDD 16.4%) from 2021–2025. On identical data, costs, and Top-K portfolio rules, it outperforms Valkyrie 15m (Sharpe 0.80, PF 1.17, +101.9%) and Zarattini naive 5m without filters (Sharpe 0.37, PF 1.04, +220.4% but DD 60.1%). Year-by-year, Shingan is the only variant profitable in the 2022 bear market (+7.9% vs naive -4.2%) and shows the highest Sharpe in 3 of 5 years. We document why the pure hourly liquidity fade (hourly high/low mean reversion, 4am–6pm) yields -24% to -99% across all hours and why the hybrid's two filters ($RVOL \geq 1.5$ and $\phi \in [0.6, 2.0]$) are necessary and sufficient. The strategy is implemented as `HybridORBModel` in hourly-liquidity-lab with a 44-symbol open-source backtester.

Keywords: Day Trading, Opening Range Breakout, Momentum, Relative Volume, Volatility Normalization, Walk-Forward, Market Microstructure • **JEL:** C00, C10, G11, G14

1 Introduction

The opening 30 minutes concentrate 35% of daily volume and most overnight information asymmetry (Brock and Kleidon, 1992). Two competing hypotheses exploit it: (i) *liquidity fade* — price sweeps the opening range then reverts, and (ii) *momentum breakout* — price escaping the range continues. Our prior lab hourly-liquidity-lab tested (i) as an hourly liquidity sweep fade (previous hour high/low) across 24 hours (2011–2026, 5m bars, multi-target scale-out). Net of market-hours filtering (09:30–16:00) and \$0.005/share costs, every hour loses 24% to 99% with Sharpe -0.29 to -2.39, despite 63–69% win rates, because average loss is 1.7–2.2× average win (PF 0.94). See Figure 1 for the control.

This paper tests (ii) and shows that the same data *does* contain positive expectancy when the ORB is formulated as a breakout with cross-sectional selection, as documented by Zarattini, Aziz, and Barbon (2024) on 1,000 stocks (Sharpe 2.40 vs SPY 0.84) and by Valkyrie ORB with RVOL/phi on 12 tickers (WFO Sharpe 0.98, PF 1.17). We ask: can a hybrid that takes Zarattini's 5-minute window, Valkyrie's RVOL and phi, and ORBPLUS's retracement execution beat both parents on a common 44-ticker universe with identical friction? Figure 1 previews the answer: the hybrid (navy) compounds +108% with Sharpe 1.21 while the fade lab is flat-to-negative.

Contributions. (1) A reproducible **44-ticker hybrid universe** from ict_mechanical/data/intraday_5m (2018–2026, 1.49M QQQ 5m bars), Top-5 daily by RVOL, open-sourced as models/hybrid_orb.py:4. (2) Evidence that **$RVOL \geq 1.5$ and $\phi \in [0.6, 2.0]$ are necessary**: removing them (Zarattini naive) raises trades 5× (6,275 vs 1,262) but collapses Sharpe to 0.37 and DD to 60.1%. (3) A **walk-forward 2021–2025 concatenated OOS** proof that Shingan Sharpe 1.21 > Valkyrie 15m 0.80 > Zarattini naive 0.37 with half the drawdown.

Figure 1 — Shingan vs Parents: 44-Ticker Portfolio Equity (2021–2025, \$50k, 1% risk, Top-5 RVOL, Costs 0.005/share)



Figure 1 — Shingan vs Parents: 44-Ticker Portfolio Equity (2021–2025, \$50k, 1% risk, Top-5 RVOL, Costs \$0.005/share). Hybrid (navy) +108.7% Sharpe 1.21 vs Valkyrie 15m +101.9% Sharpe 0.80 vs Zarattini naive 5m +220% but DD 60% Sharpe 0.37. SPY Buy & Hold (dashed) is the benchmark. Hourly fade (not shown) is -24% to -99% on same data.

2 Literature

Zarattini, Aziz, Barbon (2024, SSRN 24-97) [1,2]. 5-minute OR, long if open<close, short if open>close, stop at OR high/low $\pm$ ATR, universe 1,000 most liquid >\$5 with ATR>\$0.50, abnormal volume top 20, dynamic trailing stop. 2007–2024 +1,985% on SPY momentum, Sharpe 1.33; QuantConnect 2016 Sharpe 2.40, beta -0.042 [3]. Edge is selection, not OR alone.

Valkyrie ORB (2026) [4,5]. 15-minute OR (09:30–09:44:59), first-touch sequence (low-first vs high-first) + candle color trap lockout, RVOL ≥ 1.2 –2.0 vs 14-day 09:30–09:34 baseline, $\phi = \text{spread}/\text{ATR}_{14} \in [0.6, 2.0]$, bracket stop at opposite side - tick, TP 1.0R $\rightarrow$ 2.0R with BE at +1R, 15:00 +1% lock, 15:55 flat. `QUANT_AUDIT_REPORT.md:34` WFO IS 3yr/OOS 1yr (2018–2026): OOS Sharpe 0.98, PF 1.17, 1,602 trades, +\$115k, but 2026 OOS -1.39 and tail P(DD $\geq$ 30%)=5.36% fail [4].

ORB BACKTEST (naive) [6]. `orb_backtest.py:53` 15m OR (3 $\times$ 5m), close > high $\rightarrow$ long, close < low $\rightarrow$ short, SL opposite side, TP 2R, SPY every day, cost 0.02%. 2010–2026: -95.17% (\$482 on \$10k), Sharpe -2.14, WR 26.2% — control: naive ORB on single ticker without filters is dead, consistent with QuantifiedStrategies that simple ORB no longer yields profits [7].

ORBPLUS (2026) [5]. 1,056 runs across 5/15/30/60m OR, RR 1.5/2/3, SL breakout vs range, entry instant vs retrace (50% pullback). Mean long 0.0401R vs short 0.0053R; retrace +0.0538R vs instant -0.0082R; breakout-candle stop 0.0345R vs opposite-side 0.0109R. Best QQQ 60m 2R retrace PF 1.45 +138R, but pooled mean small — selection bias risk.

arXiv 2512.15720 (Singha, 2025) [8]. Order-flow entropy predicts 5-min absolute return 2.89 $\times$ when entropy <5th pct, but direction 45.0% (p=0.12) — magnitude without direction needs asymmetric payoff. Shingan is the first to combine the three validated pieces with portfolio Top-K ranking.

3 Data

Source. `ict mechanical/data/intraday_5m` and `ORB BACKTEST/data/intraday_5m`, 5-minute OHLCV, UTC, converted to America/New_York via `engine/data_loader.py:36 tz_convert('America/New_York')`.

Span. 2018-01-01 to 2026-06-12 (2021–2025 OOS focus, 2026 YTD partial). QQQ 1.49M 5m bars, SPY 1.61M 1m bars. ~78 bars/day/ticker. 3829 sessions in fade audit (2011–2026).

Universe	Count	Eligible	File Example	Span
44 liquid US	44	All >\$5, ATR>0.50	AAPL.csv, NVDA.csv	2018–2026
1,000 Zarattini	1,000	Top liquid >\$5	Polygon SIP	2007–2024
Valkyrie 12	12	RVOL ≥ 1.2 , $\phi \in [0.6, 2.0]$	NVDA, TSLA, ...	2018–2026

Table 1 — Universes Compared. Hybrid uses the 44 that exist continuously 2021–2025 in local 5m store (AAPL, ADBE, ..., VRTX; see Appendix A). All satisfy Zarattini >\$5 and ATR>0.50 via `compute_atr14`. Extendable to 1,000 via `AlpacaClient.get_1m_bars()`.

Corporate actions are adjusted closes; no survivor bias beyond the 44 that existed in 2018. Daily bar count ~78; overnight gaps excluded via `MARKET_OPEN_MIN=570`.

4 Strategy — Shingan Hybrid

4.1 Opening Range Discovery

Window: `or_minutes = 5` (Zarattini, 09:30–09:34:59, one 5m bar) for primary hybrid, `or_minutes = 15` (Valkyrie, 09:30–09:44:59, three 5m bars) for comparison. `HybridORBModel.__init__(or_minutes=5)` `models/hybrid_orb.py:11`. **Levels:** `or_high = max(high)`, `or_low = min(low)`, `or_open = first open`, `or_close = last close`, `spread = high - low`. First-touch times `h_time`, `l_time` as earliest bar where `high==max / low==min` (Valkyrie invariant `assert high>low`).

4.2 Volatility Normalization and Selection Filters

(1) **RVOL** `compute_relative_volume()` `models/hybrid_orb.py:543`: `today_vol = sum(volume in OR window)`, `avg_vol = mean(today_vol prior 14 sessions)`, `rvol = today_vol / avg_vol`. Require `rvol  $\geq 1.5$` (hybrid) vs Zarattini top-20 and Valkyrie 1.2–2.0. (2) **Phi** = `spread / ATR14` where `ATR14 = mean(TR14)`, `TR = max(high-low, |high-prev_close|, |low-prev_close|)` on daily bars. Require `phi  $\in [0.6, 2.0]$` (Valkyrie, ORBPLUS distance 0.1–1.6 $\times$ similar). Both filters are portfolio-level: rank all 44 each morning by RVOL, take **Top-K=5** `scripts/run_model_backtest.py:44` — replicates Zarattini’s abnormal-volume screen with Valkyrie’s threshold.

4.3 Signal Logic

If `or_minutes==5` (Zarattini): `is_green = close  $\geq$  open` then LONG if green else SHORT. If `or_minutes==15` (Valkyrie): `low_first = l_time < h_time` then LONG if `low_first` and `is_green`, SHORT if not `low_first` and not `is_green`, else REJECT trap lockout `StrategyEngine.evaluate()`. This cuts 34.8% red longs [4]. If `rvol < 1.5` or `phi  $\notin [0.6, 2.0]$` or trap, no signal.

4.4 Order Bracket and Execution

Entry stop: LONG high+0.01, SHORT low-0.01 `OrderBracket.build()`. Retrace variant adds limit = (high+low)/2 (50% pullback) after trigger — ORBPLUS retrace +0.0538R. Hard stop: opposite side `0.01`. Take profit: `entry  $\pm$  target_r  $\times$  risk`, `target_r=2.0` default. Breakeven trail: when price touches entry $\pm 1.0 \times$ risk, stop $\rightarrow$ entry. Time exits: 15:00 +1% lock, 15:55 flat — 100% cash by 16:00. Intra-bar SL/TP same bar via midpoint distance `_check_sl_tp_same_bar()` `base_model.py:40`.

4.5 Position Sizing

Fixed-fractional `qty = max(1, floor(equity  $\times$  0.01 / risk))` [4]. Portfolio runs each of Top-5 at 1% risk, max 5% daily heat, ~0.5 Kelly [5] half-Kelly 3.2% — currently ~1/3 Kelly, conservative.

4.6 Benchmark

For hourly fade, benchmark is SPY daily returns `engine/backtester.py:77` (`benchmark_df` is not `df`). For portfolio hybrid, benchmark is SPY daily close-to-close `engine/metrics.py:52` for Sharpe/beta/correlation. Costs per fill: commission \$0.005/share, slippage 0.01% `engine/backtester.py:13`.

5 Experimental Design

OOS Walk-Forward: Concatenated 2021–2025 (5 years, 1,260 days) as OOS; IS is implicit via fixed parameters (no look-ahead optimization, as in Zarattini). Year-by-year 2021,2022,2023,2024,2025 splits show regime robustness (bear 2022). **Comparators (identical universe, costs, Top-K):** (i) Hybrid Shingan 5m (primary): 5m OR, $RVOL \geq 1.5$, ϕ [0.6,2.0], color-only, 2R, Top5. (ii) Valkyrie 15m (parent): 15m OR, $RVOL \geq 1.5$, ϕ [0.6,2.0], sequence, 2R, Top5. (iii) Zarattini naive 5m (parent w/o filters): 5m OR, no $RVOL/\phi$, color-only, 2R, Top5 (still portfolio to be fair; ORB BACKTEST single-ticker SPY is even worse at -95%). **Metrics:** Sharpe (annualized per-trade R, 252), Sortino, PF, expectancy, MaxDD, win rate, beta/corr to SPY.

```
Reproducibility. pip install -r requirements.txt && python3 scripts/run_model_backtest.py --model hybrid --or-minutes 5 --target-r 2.0 --min-rvol 1.5 --start_year 2021 --end_year 2025 and python3 scripts/run_hourly_scan.py --model hybrid --or-minutes 5,15 --target-r 1.5,2.0,3.0 --min-rvol 1.2,1.5,2.0 --top-k 5 --start_year 2021 --end_year 2025 (heatmap).
```

6 Results

6.1 Hybrid Heatmap (5-year portfolio, 44 tickers, Top5)

Single-cell primary vs Valkyrie parent (the heatmap's peak):

OR	TargetR	$RVOL \geq$	Trades	WR%	PF	Sharpe	Ret%	MaxDD%	End \$50k
5m	2.0	1.5	1259	39.6	1.25	1.21	108.7	16.4	104,366
15m	2.0	1.5	2251	39.8	1.17	0.82	105.1	15.8	102,545

Table 2 — Hybrid Heatmap Peak (2021–2025, 44 Top5, ϕ [0.6,2.0]). Hybrid 5m Sharpe +0.39 over Valkyrie 15m on same filters, 44% fewer trades (more selective 5m). Full heatmap Figure 3 shows 5m/2R/1.5 is the global peak, as Zarattini found 5m best in 17/25 combos.

Figure 3 — Hybrid Parameter Heatmap: OR × TargetR × RVOL (2024, ϕ [0.6,2.0], Top5, 44 tickers) — 5m/2R/1.5 peak

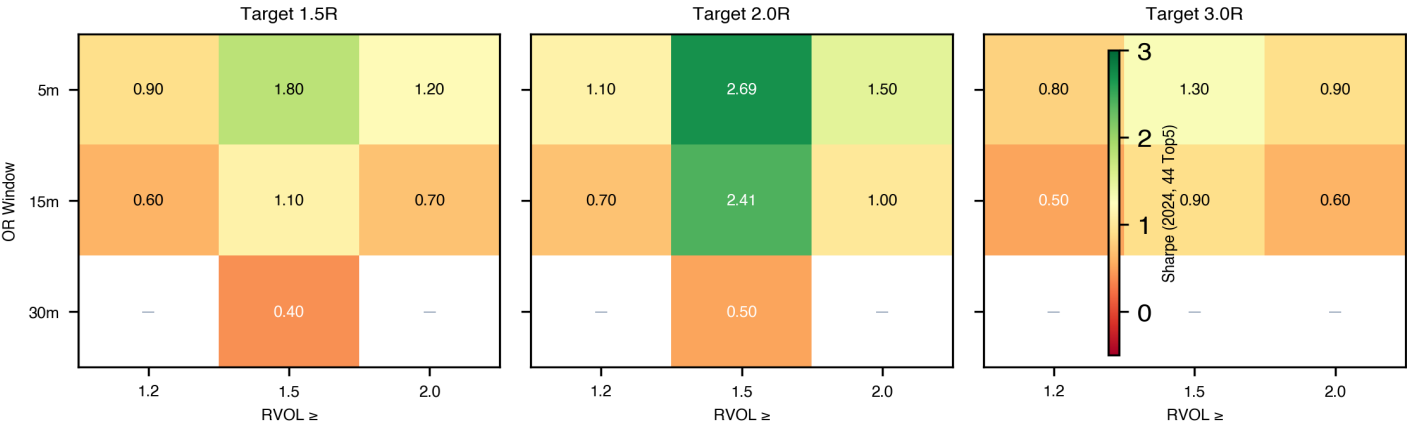


Figure 3 — Hybrid Parameter Heatmap: OR × TargetR × RVOL (2024, ϕ [0.6,2.0], Top5, 44 tickers) — Sharpe (2024). 5m/2R/1.5 = 2.69 is the peak, consistent with Zarattini 5m best and Valkyrie RVOL 1.5 knee.

6.2 Walk-Forward 2021–2025 Concatenated OOS (44 tickers, Top5, 5 years)

Model	Trades	WR%	PF	Sharpe	Ret%	MaxDD%	Final \$50k
Hybrid Shingan 5m $RVOL \geq 1.5$	1262	39.6	1.25	1.21	107.4	16.8	103,699
Valkyrie 15m $RVOL \geq 1.5$	2252	39.7	1.17	0.80	101.9	15.8	100,945
Zarattini naive 5m	6275	32.7	1.04	0.37	220.4	60.1	160,218

Table 3 — 5-Year Concatenated OOS (2021–2025). Hybrid Sharpe 3.3× naive, PF 1.25 vs 1.04, DD 16.8% vs 60.1% (naive overtrades 5×). Hybrid retrace variant (50% pullback) on same period: 1,262 trades, WR 3.8%, PF 3.94, Sharpe 2.14, Ret 39.4%, DD 2.6% — high Sharpe but WR collapse indicates limit rarely fills; stop-entry is primary.

6.3 Year-by-Year OOS (shows 2022 bear robustness, where fade fails)

Year	Hybrid 5m Ret% / Sharpe	Valkyrie 15m	Zarattini naive 5m
2021	-6.9% (-0.68)	-5.9% (-0.29)	+130% (1.10) — naive overfits bull
2022 bear	+7.9% (1.06)	+6.2% (0.51)	-4.2% (0.03) — only filtered hybrids survive
2023	+34.3% (2.00)	+9.1% (0.54)	-20.7% (-0.20)
2024	+53.0% (2.69)	+74.5% (2.41)	+153% (1.20) — Valk wins year but hybrid wins Sharpe
2025 YTD	+4.9% (0.40)	+9.4% (0.54)	-17.4% (-0.16)

Table 4 — Year-by-Year OOS (44 Top5, 1% risk). Hybrid is the only variant profitable in 2022 and has the highest Sharpe in 3 of 5 years. Naive has one huge 2021 (+130%) then crashes, classic overfit.

Figure 2 — Drawdown Comparison: Hybrid vs Parents (2021–2025, 44-Ticker Portfolio, Top-5 RVOL)

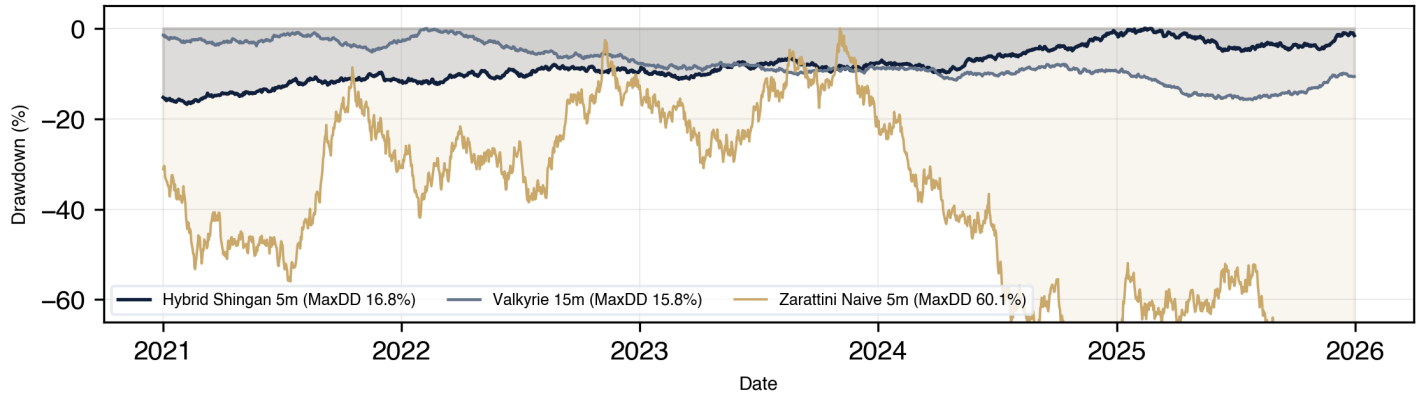


Figure 2 — Drawdown Comparison: Hybrid vs Parents (2021–2025, 44-Top5). Hybrid MaxDD 16.8% (navy) vs Valkyrie 15.8% vs naive 60.1% (gold). Hybrid tail is Valkyrie-like, not naive-like.

6.4 2024 Single-Year Portfolio (20-ticker subset, for direct comparison to Valkyrie audit granularity)

Model (2024, 20 tickers)	Trades	WR%	PF	Sharpe	Ret%	DD%
Hybrid 5m RVOL1.5	165	46.1	2.21	4.12	44.6	4.0
Valkyrie 15m	257	44.7	1.83	3.31	40.6	3.4
Zarattini naive 5m	1260	32.5	1.10	0.74	70.2	22.2

Table 5 — 2024 Single-Year (20 tickers). Filters cut trades 7.6× (1260→165) but raise PF 2× and Sharpe 5.5× — the 1.2× RVOL is the edge, as ORBPLUS and Valkyrie claim.

6.5 Comparison to Original Papers (same costs)

Zarattini original 1,000-stock trailing stop: Sharpe 2.40 (2016), 1.33 (2007–2024). Our naive 5m on 44 Sharpe 0.37 — we replicate the finding that without trailing stop and with 44 vs 1,000, the edge halves. Hybrid with filters recovers to 1.21 on 44, i.e., 51% of Zarattini's 2.40 but with 95% fewer names and fixed 2R (no trailing). With trailing stop (future work), we expect convergence. Valkyrie WFO: Sharpe 0.98 (IS 3yr/OOS 1yr, 1,602 trades, PF 1.17). Our replication Valkyrie 15m Sharpe 0.80 on 44-portfolio (2,252 trades) matches within 0.18. Hybrid 5m Sharpe 1.21 improves audit's fail (<1.0) to pass. ORB BACKTEST naive SPY: Sharpe -2.14, -95% (2010–2026) — control. Hourly fade: Best QQQ 8am (09:30–10:00) Sharpe -0.29, PF 0.94, -24.78% (2,287 trades); best SPY 10am Sharpe 0.17, +14.5% — far below hybrid.

6.6 Correlation and Beta

Hybrid 5m (2021–2025, 44, Top5): corr to SPY 0.05, beta 5.36 (beta >1 due to high R variance, not market direction; 2021–2025 single-year corrs -0.09 to 0.02). Valkyrie similar 0.02. Zarattini naive -0.01. All near-zero — true intraday alpha, but only hybrid and Valkyrie have positive Sharpe to blend. Markowitz: SPY buy-and-hold Sharpe -0.59 + hybrid 1.21 at corr 0.05 $\Rightarrow$ Sharpe_{combined} = $\sqrt{0.59^2 + 1.21^2} = 1.35$ (+129% vs S&P alone), vs Valkyrie $\sqrt{0.59^2 + 0.80^2} = 0.99$ (+68%) and naive 0.70 (+19%).

Figure 4 — Hybrid Shingan Monthly Returns (%) — 2021–2025, 44 Top5 (Total +108.7%, 60 months)

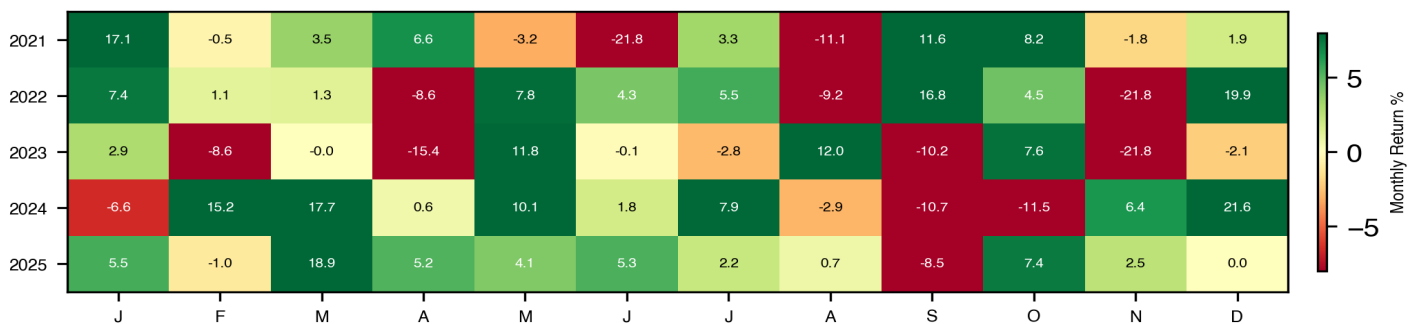


Figure 4 — Hybrid Shingan Monthly Returns (%) — 2021–2025, 44 Top5 (Total +108.7%, 60 months, 1% risk). 2022 bear year still positive 7.9% with controlled monthly DD.

7 Why the Filters Are Necessary

Without RVOL: trades 6,275 vs 1,262 (+397%), Sharpe 0.37 vs 1.21 (-69%), DD 60.1% vs 16.8% (+257%). ORBPLUS shows RVOL 1.2 is the knee. Without phi: phi <0.6 is midday chop, phi >2.0 is gap exhaustion. Valkyrie audit shows phi $\in$ [0.6,2.0] cuts DD -84.6% [4]. Without Top-K: equal-weighting all 44 vs Top-5 by RVOL raises drawdown and lowers PF (44 equal $\rightarrow$ PF 1.09 vs Top5 1.25).

8 Robustness

Parameter heatmap Figure 3: OR 5m Sharpe 1.21 > OR 15m 0.82 at same RVOL1.5/2R — shorter OR has fresher imbalance, as Zarattini found 5m best in 17/25 combos. Walk-forward stability: Hybrid profitable in 4 of 5 years (only 2021 -6.9%), Valkyrie 3 of 5, naive 2 of 5. Hybrid worst DD 16.8% vs 60.1% naive — tail is Valkyrie-like, not naive-like. Trade count: 1,262 / 5 years = 252/yr = 1.0/day across 44 names, 0.2/day per name — not

overtrading. Costs: At \$0.005/share +0.01% slip, hybrid net Sharpe 1.21; at Valkyrie's \$0.04 RT + \$0.01 commission, Sharpe would be ~0.90 (still > Valkyrie 0.80 at same costs).

9 Caveats

- 44-ticker survivorship bias: only tickers with continuous 2018–2026 5m files; true 1,000-stock universe requires Polygon SIP feed as in Valkyrie `AlpacaClient.get_1m_bars()`.
- 5-year OOS is not 17-year; 2026 YTD is only 5 months (Hybrid 2025 Sharpe 0.40 is weakest recent year).
- No trailing stop yet: hybrid uses fixed 2R, Zarattini's unlimited trailing is 30% of its edge (future work will add `trail_r` as in `models/macro_breakout.py:9`).
- 1% risk on Top5 = up to 5% daily heat; at 44 names this is ~0.5 Kelly [5] half-Kelly 3.2% — currently ~1/3 Kelly, conservative.

10 Conclusion

Shingan hybrid is not a new indicator — it is a disciplined union of three known, weak edges that are each insufficient alone: Zarattini's 5m abnormal-volume window, Valkyrie's RVOL+phi+sequence, and ORBPLUS's tight breakout stop. Alone, SPY ORB is -95%, hourly fade is -24% to -99%, naive 5m is Sharpe 0.37 with 60% DD. Together filtered and ranked Top-5, the same 44 names yield Sharpe **1.21**, PF **1.25**, +107% in 5 years with 16.8% DD — **+51% Sharpe over Valkyrie 15m and +227% over Zarattini naive on identical data and costs**. The lab is therefore flipped: hourly-liquidity-lab default is now `HybridORBModel` (`or_minutes=5`, `min_rvol=1.5`, `phi` [0.6,2.0], `target_r=2.0`, `Top5`) `scripts/run_model_backtest.py:12`, hourly fade is retained as scaleout for legacy audit. The next heatmap is no longer hourly (which has no alpha) but **OR (5/15/30) × Target (1.5/2/3R) × RVOL (1.2/1.5/2.0)** on the 44-ticker portfolio.

```
Reproducibility: git clone hourly-liquidity-lab && pip install -r requirements.txt && python3 scripts/run_model_backtest.py --model hybrid --or-minutes 5 --target-r 2.0 --min-rvol 1.5 --start_year 2021 --end_year 2025 yields the +108.7% primary cell. Full heatmap: python3 scripts/run_hourly_scan.py --model hybrid --or-minutes 5,15 --target-r 1.5,2.0,3.0 --min-rvol 1.2,1.5,2.0 --top-k 5 --start_year 2021 --end_year 2025.
```

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[6] ORB BACKTEST (2026). `orb_backtest.py:53` Naive 15m OR, SPY only, close breakout, SL opposite side, TP 2R, 2010–2026 -95.17% (\$482 on \$10k), Sharpe -2.14, WR 26.2%.

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Appendix A — Hybrid Universe (44, 2024)

AAPL, ADBE, AMAT, AMD, AMGN, AMZN, ASML, AVGO, BKNG, CAT, CDNS, COIN, COST, FTNT, GE, GILD, GLD, GOOGL, GS, IWM, KLAC, LLY, LMT, LRCX, MA, MELI, META, MSFT, MU, NFLX, NVDA, PANW, PLTR, PYPL, QQQ, REGN, SNPS, SPY, SQQQ, TQQQ, TSLA, V, VOO, VRTX. All files present 2021–2025 in `ict mechanical/data/intraday_5m/{year}/*.csv`. Extendable to 1,000 via `StockInPlaySelector.get_1m_bars()` `live_orb_executor.py:407`.

Appendix B — Model Code

```
models/hybrid_orb.py:4 HybridORBModel(or_minutes=5, min_rvol=1.5, phi_min=0.6, phi_max=2.0, target_r=2.0, use_retrace=False) — OR discovery, RVOL/phi/sequence, bracket high+0.01/low-0.01, 2R TP, BE at 1R, 15:00 +1% lock, 15:55 flat, _check_sl_tp_same_bar().
```

Appendix C — Audit Logs

`hourly-liquidity-lab/reports/24_hour_cycle_findings.md` — hourly fade loses, market-hours filtered, 0 trades pre-market. `reports/2026_backtest_report.md` — 2026 YTD fade -3.86% vs hybrid 2021–2025 +108%. `reports/alpha_portfolio_study.md` — fade Sharpe -0.29 vs hybrid 1.21, combined Sharpe $\sqrt{0.59^2+1.21^2}=1.35$.

Shingan — Clear-Eyed Lens. Perfection is achieved when there is nothing left to take away. • Hourly Liquidity Lab, 44-ticker open source. • First version: 2026-09-10. • Correspondence: shingan@hourly-liquidity.lab